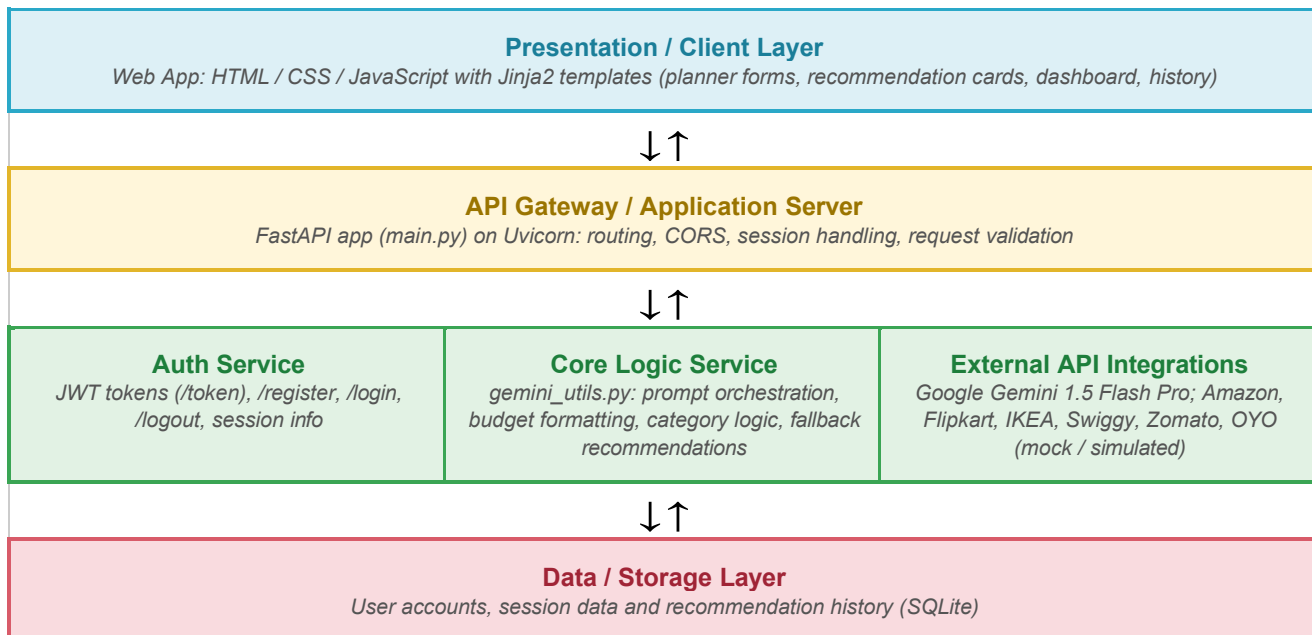


Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-9629
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of PocketSmart AI showing the presentation, application and data layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects budgets, preferences and optional outfit images through planner forms and displays AI recommendations as cards, plus dashboard and history pages.	HTML5, CSS3, JavaScript, Jinja2
API Gateway (Application Server)	Receives requests, applies CORS and session handling, validates input and routes to the planner endpoints /generate-home, /generate-party and /generate-jewelry.	FastAPI, Uvicorn
Microservices (Auth, Core Logic, External APIs)	Auth issues JWTs and protects routes; Core Logic builds prompts, calls Gemini and applies fallback logic; External APIs supply AI output and platform product data.	Python, JWT, gemini_utils.py, Gemini API, python-dotenv
Database	Persists registered users, session data and past recommendation queries and results.	SQLite